

Gaurav Mudbhatkal

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[GitHub](#) | [LinkedIn](#) | [Website](#)

OVERVIEW

Engineer specializing in multi-agent LLM systems, semantic retrieval, fine-tuning pipelines, and real-time inference. Building systems where large language models can reason, act, and make decisions over complex information environments.

TECHNICAL SKILLS

Languages and Tools: Python, SQL, Cypher, Git, FastAPI, Docker, Kubernetes, ArgoCD

Data Science and ML: PyTorch, TensorFlow, Scikit-Learn, OpenAI, LangChain, LangGraph

Big Data and Cloud: PostgreSQL, Neo4j, Elastic, AWS, Spark, Azure

Project Management: JIRA, Trello, Confluence, Notion, ServiceNow

EXPERIENCE

Data Scientist

June 2022 – Present

Limbik

Kitchener, ON

- Created end-to-end MLOps infrastructure for LLM deployment with FastAPI microservices, Kubernetes orchestration, Metaflow versioning, and real-time monitoring
- Built memory-enabled autonomous trading agents using LangGraph and OpenAI, integrating asynchronous data pipelines for real-time decision-making at scale
- Developed multi-modal retrieval system with LLM-powered query classification and vector embeddings for semantic search across trading decisions, news analysis, and agent behavior patterns
- Engineered and calibrated an LLM-based synthetic survey system generating persona-conditioned brand perception data at scale, preserving semantic fidelity while aligning outputs to real survey distributions
- Built offline evaluation pipelines using MAE, Top-K overlap, and embedding-based semantic metrics to diagnose over-selection, homogenization, and brand drift, achieving significant gains in Top-5 alignment
- Architected high-performance PostgreSQL database with optimized indexing, achieving 30% faster retrieval while scaling to 200K+ monthly survey responses with sub-second latency

Machine Learning Engineer Intern

Sep 2021 – Dec 2021

Thomson Reuters - Legal Tracker

Toronto, ON

- Optimized document classification models in **AzureML** through department feedback, improving model reliability
- Developed deployment pipelines integrating ML services with **Azure DevOps**, streamlining release processes

Data Engineer Co-op

Jan 2021 – Apr 2021

Manulife - Canada Data Office

Waterloo, ON

- Enhanced ETL workflows in Azure Data Lake using **Spark**, improving scalability and data accessibility
- Led migration of 30+ repositories to GitHub, coordinating across teams with ServiceNow, JIRA, and Confluence

Data Scientist Co-op

Sep 2019 – Dec 2019

RBC - Anti-Money Laundering

Toronto, ON

- Designed NLP pipeline for 10K+ documents using **Doc2Vec** embeddings, with entity recognition and topic modeling to extract structured insights
- Achieved 96% accuracy and 82% precision for multi-label document classification model with **Keras**

PROJECTS

Energy Swap AI | *FastAPI, Solana, React, Python*

Jan 2025

- Decentralized application for energy trading on **Solana**, integrating AI agents to monitor and optimize usage
- Built a scalable **FastAPI** backend to support dynamic trading, real-time recommendations, and AI-driven insights
- Recognized with 3rd place at the AI Agents Waterloo Hackathon among 10 competing teams

BESST (Battery Energy Storage System Toolkit) | *Flask, Gurobi, Plotly, Heroku*

Apr 2022

- Web-based decision support system to optimize residential power usage, leveraging Battery Energy Storage
- Developed a mathematical optimization model using Gurobi to minimize energy costs and GHG emissions
- Visualized cost savings and reduction in carbon footprint based on user inputs dynamically with Plotly

EDUCATION

University of Waterloo

Aug 2017 – Apr 2022

Honors, Bachelor of Applied Sciences in Management Engineering, Computing Option

Waterloo, ON